21/4/25

Polynomials

Term	Degree
1> Monomial	1 > linear poly. Ex: $ax+b, a\neq 0$
2> binomial	2 > quadratic poly. Ex: $ax^2+bx+c, a\neq 0$
3> Trinomial	3 > Ex: ax^3+bx^2+cx+d
4> Polynomial	

Introductory Questions

1 Give one example of constant polynomial =>

2 Degree of constant polynomial is 7

3 Degree of zero polynomial is Not defined

4 Write the standard form of quadratic polynomial

5 Write a polynomial of degree 50

Content of the chapter

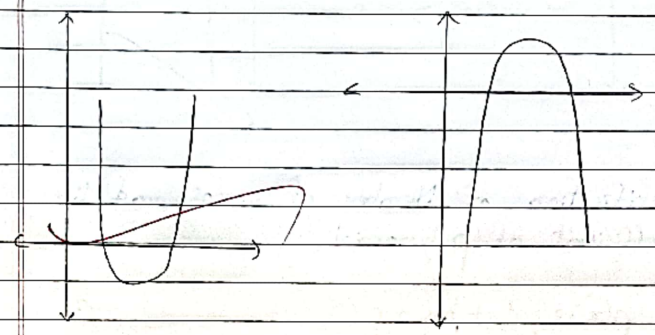
- Geometrical meaning of zeroes of polynomial.
- Relationship between zeroes of polynomial and coefficient of polynomial.

Vocabulary

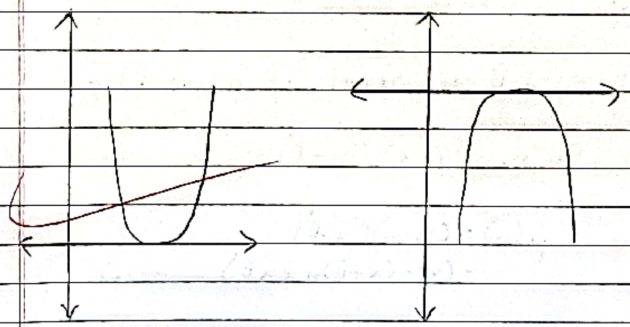
Parabola: This is a graphical representation of quadratic polynomial.

- Standard form of quadratic polynomial:

(i) $a>0$ $a<0$



(ii) $a>0$ $a<0$



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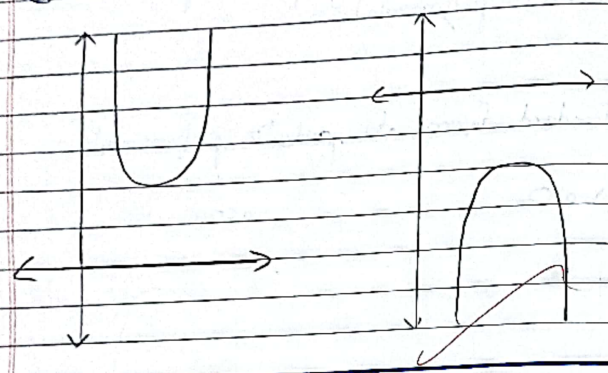
(iii) $a > 0$ $a < 0$



Verification of Number of zeroes and the coefficient of polynomial.

$$p(x) = \underbrace{ax^2}_{\text{coefficient of } x^2} + \underbrace{bx}_{\text{coefficient of } x} + \underbrace{c}_{\text{constant term}} \quad -(1)$$

Let α & β be zeroes of polynomial $p(x)$

factors $= (x-\alpha)(x-\beta)$

$$p(x) = (x-\alpha)(x-\beta)$$
$$= (x^2 - (\alpha+\beta)x + \alpha\beta) \quad -(2)$$

$$p(x) = ax^2 + bx + c$$
$$= x^2 + \frac{b}{a}x + \frac{c}{a}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = x^2 - (\alpha+\beta)x + \alpha\beta \quad (\text{from eq. (1) \& (2)})$$

By comparing,

$$-(\alpha-\beta) = a$$

$$\alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

$$\alpha+\beta = -\frac{b}{a} = \frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

Ex-2.1

i-Ans Since, the graph of $y = p(x)$ does not intersect the x-axis,
$\therefore$ The polynomial $p(x)$ has no zero

ii-Ans Since, the graph of $y = p(x)$ intersects the x-axis at one and only one point,
$\therefore$ The polynomial $p(x)$ has one and only one real zero.

iii-Ans Since, the graph of $y = p(x)$ intersects the x-axis at exact three points,
$\therefore$ The polynomial $p(x)$ has three real zeroes.

iv-Ans Since, the graph of $y = p(x)$ intersects the x-axis at two points
$\therefore$ The polynomial $p(x)$ has two for real ~~zero~~ zeroes.

v-Ans Since, the graph of $y = p(x)$ intersects the x-axis at four points,
$\therefore$ The polynomial $p(x)$ has four real ~~pointzeroes~~ zeroes.

vi-Ans Since, the graph of $y = p(x)$ intersects the x-axis at three points,
$\therefore$ The polynomial $p(x)$ has three real zeroes.

Reference book Questions

Page no. 2/11 Example ①, ②, ③	Page no. 2/13 Question ④	Page no. 2/12 Ex ⑥

Ex-1-Solu let α and $\frac{1}{\alpha}$ be the zeroes of $p(x)$

$\therefore$ Product of zeroes $= \alpha \cdot \frac{1}{\alpha} = \frac{\text{constant term}}{\text{coefficient of } x^2}$

$$1 = \frac{-(k-2)}{6}$$

$$\Rightarrow 6 = -k + 2$$

Hence, $k = -4$

Ex-3-Solu Since α and β are zeroes of $p(x) = x^2 + x - 2$

$$\alpha + \beta = \frac{-1}{1} = -1 \quad \text{and} \quad \alpha\beta = \frac{-2}{1} = -2$$

Now $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$$= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta}$$

$$= \frac{(-1)^2 - 2(-2)}{-2} = \frac{1+4}{-2}$$

Hence, $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{5}{2}$

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Ex-6-Solu Since α and β are zeroes of a polynomial $p(x) = 6x^2 - 5x + 1$,

$\therefore \alpha + \beta = \frac{5}{6}$ and $\alpha\beta = \frac{1}{6}$

Now $S = \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$= \left(\frac{5}{6}\right)^2 - \cancel{2}^{1} \times \frac{1}{\cancel{6}_{3}} = \frac{25}{36} - \frac{1}{3}$

$= \frac{13}{36}$

and $P = \alpha^2\beta^2 = \left(\frac{1}{6}\right)^2 = \frac{1}{36}$

$\therefore$ The reqd. polynomial is $x^2 - Sx + P$

$= x^2 - \frac{13}{36} + \frac{1}{36}$

$= 36x^2 - 13x + 1$

Ex-7-Solu Here, $\alpha = \frac{-3}{2}$, $\beta = \frac{1}{3}$.

$\therefore S = \alpha + \beta = \frac{-3}{2} + \frac{1}{3} = \frac{-9+2}{6} = \frac{-7}{6}$

and $P = \alpha\beta = \left(\frac{-\cancel{3}}{2}\right) \times \left(\frac{1}{\cancel{3}}\right) = \frac{-1}{2}$

$\therefore p(x) = k(x^2 - Sx + P)$

$= k\left[x^2 - \left(\frac{-7}{6}\right)x + \left(-\frac{1}{2}\right)\right] = k\left(x^2 + \frac{7x}{6} - \frac{1}{2}\right)$

Q-9-Solu $\rightarrow \alpha + \beta = \frac{-\cancel{5}}{\cancel{5}}$

$= -1$

$\& \ \alpha\beta = \frac{1}{5}$

(i) $\rightarrow \alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$

$= (-1)^2 - 2 \times \frac{1}{5}$

$= 1 - \frac{2}{5} = \frac{5-2}{5} = \frac{3}{5}$

(ii) $\rightarrow \alpha^{-1} + \beta^{-1}$

$\rightarrow \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = -1 \div \frac{1}{5} = -5$

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Ex-2.2

1-i-Solu $\rightarrow p(x) = x^2 - 2x - 8$

$= x^2 - 4x + 2x - 8$

$= x(x-4) + 2(x-4)$

$= (x+2)(x-4)$

$\alpha = -2$

$\beta = 4$

$\therefore \alpha + \beta = \frac{(-2)}{1} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$

$\Rightarrow -2+4 = 2 = -\frac{(-2)}{1} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$

$\Rightarrow \alpha\beta = (-2)(4) = \frac{-8}{1} = \frac{\text{constant term}}{\text{coefficient of } x^2}$

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ii-Solu $\rightarrow p(x) = 4s^2 - 4s + 1 = (2s-1)^2$

$\rightarrow (2s-1)^2 = (2s-1)(2s-1)$

$\rightarrow 2s - 1 = 0$ or $2s - 1 = 0$

$s = \frac{1}{2}$ or $\frac{1}{2}$

Hence, $\frac{1}{2}$ and $\frac{1}{2}$ are zeroes of $4s^2 - 4s + 1$

Now, sum of zeroes $= \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) = 1$

$= \frac{(-4)}{4} = -\frac{\text{coeff. of } x}{\text{Coeff. of } x^2}$

and product of zeroes $= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$= \frac{\text{Constant term}}{\text{Coeff. of } x^2}$

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iii-Solu $p(x) = 6x^2 - 3 - 7x = 6x^2 - 9x + 2x - 3$

$= 3x(2x-3) + 1(2x-3)$

$= (3x+1)(2x-3)$

$\therefore 2x - 3 = 0$ or $3x + 1 = 0$

$\rightarrow$ Hence, $-\frac{1}{3}$ and $\frac{3}{2}$ are zeroes of

$6x^2 - 3 - 7x$.

Now, sum of zeroes $= \left(-\frac{1}{3}\right) + \left(\frac{3}{2}\right) = \frac{-2+9}{6}$

$= \frac{7}{6} = \frac{(-7)}{6} = -\frac{\text{coeff. of } x}{\text{coeff. of } x^2}$

and product of zeroes $= \left(-\frac{1}{3}\right) \cdot \frac{3}{2} = \frac{-3}{6}$

$= \frac{\text{Constant Term}}{\text{Coeff. of } x^2}$

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iv-Solu $p(x) = 4u^2 + 8u = 4u(u+2)$

$\rightarrow 4u(u+2) = 0$

$\rightarrow u(u+2) = 0$

$\rightarrow u = 0$ or $u = -2$

Hence, -2 and 0 are zeroes of $4u^2 + 8u$

sum of zeroes $= -2 + 0 = \frac{-2}{1} = -\frac{8}{4} = -\frac{\text{coefficient of } u}{\text{coefficient of } u^2}$

product of zeroes $= (-2)(0) = \frac{0}{4} = \frac{\text{constant term}}{\text{coefficient of } u^2}$

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v-Solu $\Rightarrow t^2 - 15 = (t-\sqrt{15})(t+\sqrt{15})$

$\therefore t^2 - 15 = 0$

$(t-\sqrt{15})(t+\sqrt{15}) = 0$

$\Rightarrow t - \sqrt{15} = 0$ or $t + \sqrt{15} = 0$

Hence, $-\sqrt{15}$ and $\sqrt{15}$ are the zeroes of $t^2 - 15$

Sum of the zeroes $= (-\sqrt{15}) + (\sqrt{15})$

$= \frac{0}{1} = \frac{\text{coefficient of } t}{\text{coefficient of } t^2}$

Product of zeroes $= (\sqrt{15})(-\sqrt{15}) = \frac{-15}{1} = \frac{\text{constant term}}{\text{coefficient of } t^2}$

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vi-Solu $\Rightarrow 3x^2 - x - 4 = 3x^2 + 3x - 4x - 4$

$= 3x(x+1) - 4(x+1)$

$= (3x-4)(x+1)$

$\Rightarrow x+1 = 0$ or $3x - 4 = 0$

$\Rightarrow x = -1$ or $x = \frac{4}{3}$

Hence, the zeroes of $3x^2 - x - 4$ are -1 and $\frac{4}{3}$

Sum of zeroes $= -1 + \frac{4}{3} = \frac{-3+4}{3} = \frac{1}{3} = \frac{\text{coefficient of } x}{\text{coefficient of } x^2}$

product of zeroes $= (-1)(\frac{4}{3}) = -\frac{4}{3} = \frac{\text{constant term}}{\text{coeff. of } x^2}$

2-i-Solu $\Rightarrow \alpha + \beta = \frac{1}{4}$

$\Rightarrow \alpha \cdot \beta = -1$

$\Rightarrow K[x^2 - (\alpha+\beta)x + \alpha\beta]$

$\Rightarrow K[x^2 - \frac{1}{4}x - 1]$

$\Rightarrow K[\frac{4x^2 - 1x - 4}{4}]$

$\Rightarrow K[4x^2 - x - 4]$, where K is $\frac{1}{4}$

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ii-Solu $\alpha + \beta = \sqrt{2}$

$\alpha \cdot \beta = \frac{1}{3}$

$\Rightarrow K[x^2 - (\alpha+\beta)x + \alpha\beta]$

$\Rightarrow K[x^2 - \sqrt{2}x + \frac{1}{3}]$

$\Rightarrow K[\frac{3x^2 - 3\sqrt{2}x + 1}{3}]$

$\Rightarrow K[3x^2 - 3\sqrt{2}x + 1]$, where K is $\frac{1}{3}$.

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iii-Solu $\alpha + \beta = 0$

$\alpha \cdot \beta = \sqrt{5}$

$\Rightarrow K[x^2 - (\alpha+\beta)x + \alpha\beta]$

$\Rightarrow K[x^2 + \sqrt{5}]$, where K = 1.

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iv-Sol: $\alpha+\beta = 1$

$\alpha \cdot \beta = 1$

$\Rightarrow K[x^2 - (\alpha+\beta)x + \alpha\beta]$

$\Rightarrow K[x^2 - x + 1]$, where $K = 1$

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v-Sol: $\alpha+\beta = -\frac{1}{4}$

$\alpha \cdot \beta = \frac{1}{4}$

$\Rightarrow K[x^2 - (\alpha+\beta)x + \alpha\beta]$

$\Rightarrow K[x^2 + \frac{1}{4}x + \frac{1}{4}] = K\left[\frac{4x^2+x+1}{4}\right]$

$\Rightarrow K[4x^2+x+1]$, where $K = \frac{1}{4}$

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vi-Sol: $\alpha+\beta = 4$

$\alpha \cdot \beta = 1$

$\Rightarrow K(x^2 - (\alpha+\beta)x + \alpha\beta)$

$\Rightarrow K(x^2 - 4x + 1)$, where $K = 1$

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